

Quantum Thermodynamic: Time definition, Causality, Markov property, Multi-path Future

Zou Zhi Kai 邹志凯 Email: zhiyan.zou@foxmail.com Independent Researcher Shenzhen; China
<https://orcid.org/0009-0000-4279-1064>

Note: This article serves as a brief summary of previous discussions on the analyticity of quantum thermodynamics, the discrete definition of time, the multi-path nature of the future, and the Markov-property of causality.

Abstract: Since quantum thermodynamics is mentioned, the background is not a continuous field but a quantized network. Under this quantized network framework and at the Planck time scale, quantum thermodynamics is no longer statistical but analytical in nature.

At the Planck time scale, any statistical-probabilistic physical description reduces to an analytical one. In this paper, based on a quantum spacetime model at the Planck scale, time is redefined as a quantum thermodynamic phenomenon driven by the state transitions of a discrete spacetime network. We propose a novel framework in which time ceases to be a continuous parameter but emerges naturally as a discrete sequence from the entropy-increasing evolution of the Spatial Elementary Quantum (SEQ) network. This model tightly couples time with the law of entropy increase, revealing that the essence of time is a statistical outcome of state changes in the microscopic structure of the universe. Through this framework, we not only unify the perspectives of thermodynamics and quantum mechanics but also provide a new theoretical foundation for understanding the arrow of time. The model predicts non-uniformity of time at local scales and explains time dilation in relativity, offering fresh insights for the development of quantum gravity theories.

Keywords: Quantum thermodynamics; Definition of time; Discrete spacetime; Law of entropy increase; Planck scale; Gravitational time dilation; Markovian causality;

1. Spacetime Framework In this model, spacetime is not a continuous background but a dynamic network composed of discrete Space Elementary Quanta (SEQ). Each SEQ is an indivisible unit at the Planck scale, where elastic interactions and excited states encode all physical phenomena. The evolution of spacetime manifests as state transitions of the SEQ network, while time emerges naturally as an order parameter governing these transitions. Due to the discrete nature of the SEQ network, any physical process (e.g., energy transfer, particle motion) appears as the propagation of excitations between adjacent SEQs, rather than as classical "motion through space." The speed of light (c) serves as the limiting rate for excitation propagation, ensuring the preservation of relativistic causal structure.

2. Entropy Definition and Spacetime-Entropy Mapping In this framework, entropy is defined as the multiplicative accumulation of SEQ energy distributions, rather than the traditional Boltzmann statistical form. Specifically, the entropy at a given moment is determined by the product of the energies of all SEQs involved in state transitions at that instant. This multiplicative entropy is more natural in discrete systems and can revert to the classical entropy expression under logarithmic transformation. The law of entropy increase here manifests as the spontaneous

tendency of the SEQ network's energy distribution toward more uniform configurations, driving the unidirectionality of the time arrow.

The spacetime-entropy mapping is characterized as follows: Each global or local state transition of the SEQ network corresponds to an entropy value, and the non-uniform growth sequence of entropy values defines the passage of time. This mechanism not only explains the macroscopic second law of thermodynamics but also provides a microscopic foundation for time evolution at quantum scales.

3. Relativistic Effects: A Temporal Interpretation

3.1 Special Relativistic Phenomena (Time Dilation & Length Contraction) In this framework, time dilation and length contraction—key effects of special relativity—do not arise from geometric deformations of spacetime itself but rather from observational discrepancies caused by dynamic variations in light-path propagation between inertial reference frames. Since the speed of light (c) is the maximum excitation propagation rate in the SEQ network, all spacetime measurements for moving objects are constrained by the discreteness of information transfer. The specific mechanisms are as follows:

Time Dilation in SR (Lorentz-FitzGerald Contraction)

- When an observer measures a moving clock, the recorded time depends on the count of SEQ transitions along the round-trip path of light signals.
- Due to relative motion, the light path for a moving clock lengthens (as signals must "chase" the moving clock), resulting in a reduced observed frequency of SEQ transitions.
- This manifests as slower perceived time flow, purely a counting effect constrained by the invariance of c , rather than an actual change in the clock's internal physical processes.

Length Contraction (Lorentz-FitzGerald Contraction)

Spatial distance measurements also rely on light-signal propagation paths through SEQs.

The spatial extent of a moving object (quantified by SEQ count in the direction of motion) appears contracted due to dynamic adjustments in light-path geometry.

This is a projection effect caused by the finite speed of light and relative motion, not an actual compression of the object's SEQ network.

Mathematically, this framework yields a discretized version of the Lorentz transformation, consistent with textbook special relativity.

Difference from General Relativity:

Unlike in general relativity, the effects of special relativity do not involve gravity or actual deformations of the SEQ network. Instead, they solely reflect differences in observational constraints between inertial reference frames.

Physical Meaning of Lorentz Transformations:

The Lorentz transformations essentially represent discrete counting calibrations of the same SEQ excitation event across different reference frames.

Their mathematical form arises from the global constraint imposed by the invariance of the speed of light (c) on information propagation.

This makes them a measurement synchronization rule in a discrete spacetime framework, rather than a description of spacetime "stretching" or "shrinking."

Reinterpretation of Special Relativity:

This explanation fully reduces special relativity to a dynamic observational phenomenon within a discrete spacetime network.

It eliminates the abstract notion of "spacetime itself expanding or contracting" present in continuous spacetime models.

3.2 General Relativity (Gravitational Time Dilation and Equivalence Principle)

In this framework, the curvature of spacetime induced by mass or acceleration is fundamentally reinterpreted as local compression and stretching of the SEQ network. The key mechanisms are:

Gravitational Time Dilation:

- Strong gravitational fields (e.g., near black holes) compress the spacing between SEQs
- This compression reduces the frequency of state transitions in the local SEQ network
- Result: Observable time dilation effects identical to GR predictions
- Physical interpretation: Time flows slower because quantum state changes occur less frequently in compressed regions

Equivalence Principle: In the equivalence principle, acceleration, velocity, or other factors affecting the metric are essentially dynamic deformations exerted by moving objects on the SEQ network, whose effects are quasi-equivalent to those of a static gravitational field.

Theoretical Advantages:

Naturally avoids singularity problems of continuous spacetime theories
Provides a physical mechanism for "spacetime curvature" at quantum scales
Maintains all verified GR predictions while offering a discrete underlying structure
The SEQ network's discrete nature automatically provides UV cutoff for gravitational effects

The model shows how Einstein's field equations emerge as an effective description of the statistical behavior of the underlying quantum network at macroscopic scales, while providing a more fundamental microscopic explanation for gravitational phenomena.

4 Analytical Quantum Thermodynamics

In this framework, quantum superposition originates from the multi-layered, multi-axis rotation of dynamic resonance within a particle's internal SEQ structure—potentially forming a high-dimensional phase space and uncertainty prior to measurement. Eigenstates correspond to metastable solutions of this system, determined by intrinsic parameters: mass distribution (gravitational potential storage), electromagnetic coupling, chiral symmetry constraints (e.g., Higgs locking), and initial conditions. Measurement causes the rotating structure to collapse into one of the permitted states. The uncertainty of superposition arises from high-speed dynamic

rotation and resonance.

Under this framework model, all probability functions reduce to analytical functions at the Planck time scale. In other words, this model is a quasi-deterministic model at the Planck time scale. As previously discussed, although past and present states are deterministic, under the constraints of entropy increase and the principle of least action, the future still retains degrees of freedom. The future depends only on the state of the immediately preceding moment—Markovian, meaning the future is not entirely predetermined.

Setting the Planck time as the minimum time scale is equivalent to acknowledging that all probability functions reduce to analytical functions at the Planck time scale. Conversely, if one insists on the probabilistic Copenhagen interpretation of quantum mechanics at the Planck time scale, it effectively denies the Planck time as the minimum scale of cosmic evolution.

5 Causality in Quantum Thermodynamics: Entropy increase is the primary causal law of the universe. Due to the existence of isoentropic transformations and different entropy-increasing paths, even when further constrained by the principle of least action to reduce possible degrees of freedom in future state transformation paths, the possible spacetime evolution paths are not necessarily unique. This is because, within this model's framework, there may exist multiple least-action evolution paths. However, what determines the next moment in the future is the state of the immediately preceding moment - meaning that in this model, causality is Markovian, with a certain degree of freedom.

More discussion: Time-Entropy Mirroring via Space Transformation and Mass-Gravity Duality via QCD-Higgs Synergy. Preprints. <https://doi.org/10.20944/preprints202505.0270.v2>